

Analysis Notes

Analysis 1, course material overview

Eklavya Raman

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Chapter 1

The Real Field

1.1 Ordered Sets and Fields

1.1.1 Ordered Sets

We use basic set theoretic notation and definitions.

Definition 1.1.1.1

Order and Ordered Set

We define an order and an ordered set as follows:

1. An **order** on a set S is a relation $<$ defined such that:
 - (a) For all $x, y, z \in S$, either $x < y$, $y < x$ or $x = y$.
 - (b) For all $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$. (Transitivity)
2. An **ordered set** is a set S with an order $\leq$ defined on it.

Definition 1.1.1.2

Upper and Lower Bounds

Let S be an ordered set and $A \subset S$.

1. An element $x \in S$ is an **upper bound** of A if for all $a \in A$, $a \leq x$. Then A is **bounded above** if there exists an upper bound of A .
2. An element $x \in S$ is a **lower bound** of A if for all $a \in A$, $x \leq a$. Then A is **bounded below** if there exists a lower bound of A .

Theorem 1.1 (Every finite non-empty set is bounded). *Every finite non-empty subset A of an ordered set S is bounded above and below. That is, there exists an upper bound and a lower bound of A in S .*

Proof. We show this using induction. As a base case, let -

$$A_1 = \{x_1\}$$

Then, we see that x_1 is an upper bound of A_1 and also a lower bound of A_1 . Now, let us assume that for some $n \in \mathbb{N}$, the statement holds for all sets of size n . That is, for any set A_n of size n , there exists an upper bound and a lower bound. Now, let us consider a set A_{n+1} of size $n + 1$. We can write -

$$A_{n+1} = A_n \cup \{x_{n+1}\}$$

where A_n is a set of size n . By the inductive hypothesis, we know that A_n has an upper bound u and a lower bound l . Now, we consider the following cases:

1. If $x_{n+1} \leq u$, then u is an upper bound of A_{n+1} .

2. If $x_{n+1} > u$, then x_{n+1} is an upper bound of A_{n+1} .
3. If $x_{n+1} \geq l$, then l is a lower bound of A_{n+1} .
4. If $x_{n+1} < l$, then x_{n+1} is a lower bound of A_{n+1} .

Thus, in all cases, we see that A_{n+1} has an upper bound and a lower bound. Hence, by induction, we conclude that every finite set is bounded above and below. $\square$

Definition 1.1.1.3

Supremum and Infimum

Let S be an ordered set and $A \subset S$.

1. The supremum of A , denoted $\sup A$, is the least upper bound of A i.e. -

$$\sup A = \min(\{x \in S : x \text{ is an upper bound of } A\})$$

2. The infimum of A , denoted $\inf A$, is the greatest lower bound of A .

Important Points:

- The supremum and infimum of a set may not exist.
- Either of the supremum or infimum may not be an element of the set.
- If the supremum or infimum exists, it is unique.

Theorem 1.2 (Supremum and Infimum are unique). *Let S be an ordered set and $A \subset S$. If $\sup A$ exists, then it is unique. Similarly, if $\inf A$ exists, then it is unique.*

Proof. Let α and β be two supremums of A . Then, by definition, α is an upper bound of A and β is an upper bound of A . Since α is the least upper bound, we have $\alpha \leq \beta$. Similarly, since β is the least upper bound, we have $\beta \leq \alpha$. Hence, we conclude that $\alpha = \beta$. Thus, the supremum is unique. The proof for the infimum is similar. $\square$

Definition 1.1.1.4

Least upper bound property

An ordered set S is said to have the least upper bound property if every non-empty subset of S that is bounded above has a supremum in S . That is -

$$\forall A \subset S : ((A \neq \emptyset) \wedge (A \text{ is bounded above})) \implies \sup A \in S$$

Theorem 1.3 (Existence of Infimum in S with the least upper bound property). *Let S be an ordered set with the least upper bound property. Then for any non empty subset A of S that is bounded below and L be the set of all lower bounds of A , then the following hold:*

$\alpha = \sup L$ exists in S and is the infimum of A , i.e.:

$$\exists \alpha \in S : \alpha = \sup L = \inf A \implies \inf A \in S$$

Proof. Since A is bounded below, L is non-empty (there exists at least one lower bound of A). Now:

$$L = \{y \in S : \forall x \in A, y \leq x\}$$

Since S has the least upper bound property, L is bounded above (by any element of A) and hence by the property, $\sup L$ exists in S . Now, we need to show that $\sup L = \inf A$.

Let $\alpha = \sup L$. Then:

$$\gamma < \alpha \implies \gamma \text{ is not an upper bound of } L$$

Thus $\gamma \notin A$. Therefore, $\alpha \leq x$ for all $x \in A$. Hence, α is a lower bound of A .

$$\alpha < \beta \implies \beta \notin L$$

since α is an upper bound of L .

Therefore, $\alpha \in L$ but any number greater than α is not in L . Hence, α is the greatest lower bound of A . Thus, $\alpha = \inf A$. Since α is in S , we have shown that $\inf A \in S$. $\square$

1.1.2 Fields

Definition 1.1.2.1

Field

A **field** is a set F with two operations $+$ and $\cdot$ defined on it such that it satisfies the following axioms:

• **Addition Axioms:**

1. $x \in F \wedge y \in F \implies x + y \in F$ (Closure under addition)
2. $x \in F \wedge y \in F \implies y + x = x + y$ (Commutativity of addition)
3. $x \in F \wedge y \in F \wedge z \in F \implies (x + y) + z = x + (y + z)$ (Associativity of addition)
4. There exists an element $0 \in F$ such that for all $x \in F$, $x + 0 = x$ (Existence of additive identity)
5. For every $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$ (Existence of additive inverse)

• **Multiplication Axioms:**

1. $x \in F \wedge y \in F \implies x \cdot y \in F$ (Closure under multiplication)
2. $x \in F \wedge y \in F \implies y \cdot x = x \cdot y$ (Commutativity of multiplication)
3. $x \in F \wedge y \in F \wedge z \in F \implies (x \cdot y) \cdot z = x \cdot (y \cdot z)$ (Associativity of multiplication)
4. There exists an element $1 \in F$ such that for all $x \in F$, $x \cdot 1 = x$ (Existence of multiplicative identity)
5. For every $x \in F$ such that $x \neq 0$, there exists an element $x^{-1} = 1/x$ such that $x \cdot x^{-1} = 1$ (Existence of multiplicative inverse)

• **Distributive Axiom:**

$$\forall x, y, z \in F : x \cdot (y + z) = x \cdot y + x \cdot z$$

Example 1.1.2.1 -

The set of rational numbers $\mathbb{Q}$ with the usual operations of addition and multiplication is a field.

Definition 1.1.2.2

Ordered Field

An **ordered field** is a field F with an order $\leq$ defined on it such that:

1. For all $x, y, z \in F$, if $x < y$, then $x + z < y + z$ (Order is preserved under addition)
2. For all $x, y, z \in F$, if $0 < x$ and $0 < y$, then $0 < x \cdot y$ (Order is preserved under multiplication)

Theorem 1.4 (Existence Theorem of the Real Field). *There exists a unique ordered field $\mathbb{R}$ such that:*

1. $\mathbb{R}$ has the least upper bound property.
2. $\mathbb{R}$ contains the rational numbers $\mathbb{Q}$ as a subfield.

*This field is called the **real field** or the **field of real numbers**.*

Theorem 1.5 (Archimedean Property and Denseness of $\mathbb{Q}$ in $\mathbb{R}$). *The field of real numbers $\mathbb{R}$ satisfies the following properties:*

1. **Archimedean Property:** *For any two real numbers x and y , there exists a natural number n such that $nx > y$, i.e.:*

$$\forall x, y \in \mathbb{R}, \exists n \in \mathbb{N} : nx > y$$

2. **Density of $\mathbb{Q}$ in $\mathbb{R}$:** For any two real numbers x and y with $x < y$, there exists a rational number $q \in \mathbb{Q}$ such that $x < q < y$, i.e:

$$\forall x, y \in \mathbb{R}, (x < y) \implies \exists q \in \mathbb{Q} : x < q < y$$